Amharic-Centric Large Language Models: Data-Efficient Continual Pre-training and Incremental Adaptation for Low-Resource Natural Language Processing

Kindie Alebachew Tsega[1,*], **Durga Prasad Sharma**[1], **Mesay Samuel Gondere**[1], and **Mohammed Abebe Yimer**[1]

[1]Faculty of Computing and Software Engineering,
Arba Minch Institute of Technology, Arba Minch University,
Arba Minch, Ethiopia.

ORCID iDs and Email Addresses:

Kindie Alebachew Tsega: ORCID: 0009-0005-2613-3914, Email: besukind@gmail.com

Durga Prasad Sharma: ORCID: 0000-0002-1654-901X, Email: dp.shiv08@gmail.com

Mesay Samuel Gondere: ORCID: 0000-0002-5070-2727

Mohammed Abebe Yimer: ORCID: 0000-0003-0622-4841

[*]**Corresponding Author:** Kindie Alebachew Tsega (Email: besukind@gmail.com)

Abstract

Transformer-based language models face severe performance degradation during sequential adaptation on morphologically rich, low-resource languages such as Amharic. This paper investigates data-efficient continual pre-training and incremental adaptation of Amharic language models by comparing Weight-Decomposed Low-Rank Adaptation (DoRA), standard LoRA, LoRA with Elastic Weight Consolidation (LoRA+EWC), and full fine-tuning (Full-FT) on `roberta-base-amharic`. We evaluate sequential Named Entity Recognition (NER) across three random seeds and introduce **EthioEnAm-Parallel**, an open-source English–Amharic parallel corpus (19,386 verified sentence pairs from OPUS-100 and multi-domain reasoning benchmarks), complemented by an auxiliary synthetic bitext generator. In sequential NER, DoRA and LoRA achieve a mean macro-F1 of 0.746 (an 8.4 pp gain over Full-FT at 0.662) while updating only 1.2–1.3 % of parameters and demonstrating positive backward transfer (BWT $\approx +0.05$). On cross-lingual parallel adaptation, DoRA achieves 26.82 BLEU and 51.40 ChrF++, effectively eliminating catastrophic forgetting compared to Full-FT (BWT of +0.84 vs. −2.14 BLEU points). Synthetic parallel bitext improves forward transfer (+1.12 vs. +0.95) but exhibits syntactic rigidity that bounds peak fluency. The framework is positioned against the I-GANH Khasi machine translation architecture, validating parameter-efficient subspace adaptation for low-resource continual learning.

Keywords: Continual learning, catastrophic forgetting, weight-decomposed low-rank adaptation (DoRA), parameter-efficient fine-tuning (PEFT), Amharic NLP, low-resource language models, named entity recognition, English–Amharic parallel corpus, machine translation.

1 Introduction

Large language models (LLMs) have transformed natural language processing (NLP), attaining strong performance in text understanding, generation, and complex reasoning through large-scale pre-training on diverse corpora [1–3]. Despite these advances, pre-trained LLMs face a fundamental limitation: once trained, they cannot efficiently incorporate new information without costly retraining or fine-tuning, especially in dynamic environments where data distributions evolve [4].

Continual Learning (CL) addresses this by enabling models to sequentially acquire new knowledge while preserving previously learned information [5, 6]. However, integrating CL into large transformers remains a formidable challenge because gradient-based updates optimised for new data overwrite parameters encoding prior knowledge—a phenomenon known as *catastrophic forgetting* [6, 7].

These challenges are acutely amplified for Amharic, an Afro-Asiatic Semitic language spoken by over 57 million people as a first language. Amharic is written in the Ethiopic (Ge'ez) syllabary, exhibits rich morphological agglutination, and is severely under-represented in public pre-training corpora [13, 14]. Existing multilingual models underperform on Amharic relative to high-resource languages [15], and purpose-built Amharic models remain relatively small and resource-constrained [16].

Parameter-Efficient Fine-Tuning (PEFT) reduces adaptation costs by updating a small, task-relevant parameter subset while freezing the backbone [8, 9]. Among recent PEFT methods, Weight-Decomposed Low-Rank Adaptation (DoRA) [10] extends LoRA by decomposing each weight matrix into independent magnitude and directional components, yielding richer updates without increasing the trainable-parameter budget. Despite its promise, DoRA has not been systematically evaluated in continual pre-training contexts or for low-resource Semitic languages.

Research gap. CL strategies remain largely unexplored for Amharic. Existing Amharic models are static; scalable incremental adaptation pipelines are absent. Advanced PEFT techniques—in particular DoRA—have not been studied in sequential adaptation settings for low-resource languages. The integration of PEFT into *continual pre-training* (as opposed to isolated fine-tuning) is critically lacking.

Central research question. Can weight-decomposed, data-efficient continual adaptation improve Amharic language-model performance while reducing computational costs and mitigating catastrophic forgetting, without retraining the entire model from scratch?

This paper makes the following contributions:

1. We evaluate DoRA within a controlled multi-stage continual learning protocol for Amharic—the first such study for a low-resource Afro-Asiatic language.

2. We curate, format, document, and release **EthioEnAm-Parallel**, an open-source English–Amharic parallel corpus comprising 19,386 verified sentence pairs derived from OPUS-100 and localized multi-domain reasoning benchmarks, structured into sequential continual learning shards alongside an explicitly marked auxiliary synthetic parallel dataset.

3. We provide a systematic comparison of DoRA, LoRA, LoRA+EWC, and Full-FT under

identical conditions across both monolingual Amharic NER and cross-lingual English–Amharic parallel adaptation.

4. We quantify catastrophic forgetting via backward and forward transfer metrics derived from a full performance matrix.

5. We characterise the Pareto trade-off between trainable parameters, GPU memory, training time, and downstream accuracy.

6. We release reproducible experimental artefacts: data processing pipelines, synthetic generators, code, configuration files, and evaluation scripts.

7. We critically analyse and transfer methodological insights from the GAN-based incremental learning framework (I-GANH) for Khasi machine translation [37], assessing its staged protocol, stability loss, and cross-lingual bitext synthesis.

8. We audit the evaluation protocol, re-analyse the reported statistics with effect sizes and multiplicity control, and specify eight complementary experiments (E1–E8) with runnable code.

2 Background and Theoretical Foundations

2.1 Catastrophic Forgetting

Catastrophic forgetting (catastrophic interference) is the tendency of neural networks to abruptly lose previously acquired representations when trained sequentially on new data [6, 7]. In transformer-based LLMs the phenomenon is exacerbated by massive parameter spaces, since standard SGD does not constrain gradient updates to regions that preserve prior representations [4, 22].

2.2 Continual Learning Approaches

CL methods fall into three broad families:

Regularisation-based methods penalise changes to parameters important for prior tasks. Elastic Weight Consolidation (EWC) uses the diagonal Fisher Information Matrix (FIM) as an approximation of parameter importance [6]. While theoretically sound, EWC incurs $O(|\theta|)$ additional memory per task and requires a Monte-Carlo pass over each prior dataset to estimate the FIM, making it prohibitively expensive at transformer scale [23].

Replay-based methods maintain a buffer of historical samples interleaved with new data [11]. Managing representative buffers for large corpora is often impractical, and synthetic replay introduces generative complexity.

Parameter isolation/expansion methods allocate dedicated parameter subsets per task, eliminating interference at the cost of linear storage growth [8].

2.3 Low-Rank Adaptation (LoRA)

LoRA [9] constrains weight updates to a low-rank subspace. Given a pre-trained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d\times k}$, LoRA represents its update as

$$\mathbf{W} = \mathbf{W}_0 + \Delta\mathbf{W} = \mathbf{W}_0 + \mathbf{BA}, \tag{1}$$

where $\mathbf{B} \in \mathbb{R}^{d\times r}$, $\mathbf{A} \in \mathbb{R}^{r\times k}$, and rank $r \ll \min(d, k)$. Only $\mathbf{A}$ and $\mathbf{B}$ are trained; $\mathbf{W}_0$ is frozen. The update is scaled by α/r. At inference, $\mathbf{BA}$ can be merged into $\mathbf{W}_0$ with no latency penalty.

2.4 Weight-Decomposed Low-Rank Adaptation (DoRA)

DoRA [10] decomposes each pre-trained weight matrix into a *magnitude vector* $\mathbf{m} \in \mathbb{R}^{1\times k}$ and a *directional matrix* $\mathbf{V} \in \mathbb{R}^{d\times k}$:

$$\mathbf{W} = \mathbf{m} \cdot \frac{\mathbf{V}}{\|\mathbf{V}\|_c}, \tag{2}$$

where $\|\cdot\|_c$ denotes column-wise ℓ_2 normalisation. During adaptation the directional component is reparameterised via LoRA ($\mathbf{V} \leftarrow \mathbf{V}_0 + \mathbf{BA}$) while $\mathbf{m}$ is trained directly:

$$\mathbf{W}_{\text{DoRA}} = \mathbf{m} \cdot \frac{\mathbf{W}_0 + \mathbf{BA}}{\|\mathbf{W}_0 + \mathbf{BA}\|_c}. \tag{3}$$

Decoupling magnitude and direction enables richer gradient dynamics: magnitude captures global scaling; direction captures relational changes. Trainable parameters are $\mathbf{m}$, $\mathbf{B}$, and $\mathbf{A}$, adding one scalar per column of $\mathbf{W}_0$ beyond standard LoRA.

2.5 Amharic Linguistic Properties

Amharic is a VSO/SVO-variable, pro-drop, highly agglutinative language written in the Ethiopic (Ge'ez) syllabary. Key NLP challenges:

- **Morphological richness**: a single Amharic verb encodes tense, aspect, polarity, agreement, and derivational information.
- **Orthographic variation**: multiple spellings of homophonous characters require script normalisation.
- **Tokeniser coverage**: SentencePiece/BPE vocabularies trained on multilingual data over-fragment Amharic word forms, increasing sequence length [20].
- **Resource scarcity**: annotated datasets for NER, POS, QA, and summarisation are scarce.

3 Related Work

3.1 Continual Learning in LLMs

CL enables sequential knowledge acquisition without forgetting. EWC [6] established the regularisation paradigm. Recent work on CL in NLP is reviewed by Wu et al. [22] and

Wang et al. [23]. Catastrophic forgetting has been documented during continual fine-tuning of LLMs [29], and LoRA has been found to learn less but forget less than full fine-tuning [28], motivating data-efficient and parameter-efficient CL strategies.

3.2 Parameter-Efficient Fine-Tuning

PEFT reduces adaptation cost by updating a small parameter subset [8, 9, 12]. LoRA is widely adopted; DoRA extends it with weight decomposition [10]. PEFT methods have primarily been evaluated on isolated fine-tuning tasks; their behaviour under *sequential pre-training* with distributional shift remains underexplored [4, 24].

3.3 Amharic NLP

Early Amharic NLP relied on multilingual transformers such as mBERT and XLM-R [15]. AmharicBERT [16] and `roberta-base-amharic` [17, 18] provide Amharic-specific encoders. EthioLLM [13] and Walia-LLM [19] extend instruction tuning to Amharic. Text classification, NER, and news topic classification are the most studied tasks; continual pre-training and incremental adaptation remain insufficiently studied [14].

3.4 Incremental Learning for Low-Resource Languages

Incremental approaches to MT for low-resource languages include NMT domain adaptation with EWC or replay, hybrid SMT-NMT systems, and GAN-based data augmentation [25–27, 37]. These works motivate frameworks that can absorb new data without full retraining—an objective directly relevant to Amharic continual pre-training.

4 Critical Analysis of the I-GANH Reference

4.1 Paper Identity and Verification

The reference paper is: I. Syiemlieh, Z. Ahmed, A. K. Maji, and S. Das, "GAN-based Incremental Learning for English to Khasi Machine Translation: A Low-Resource Language Model," *Computación y Sistemas*, vol. 30, no. 3, pp. 1991–2007, 2026. DOI: 10.13053/CyS-30-3-6641. The PDF (file `66411406901PB.pdf`) was inspected directly; all attributions below are derived from verified content.

4.2 Research Problem and Motivation

The paper targets English-to-Khasi machine translation (MT). Khasi is a morphologically complex Austroasiatic language spoken in Meghalaya, India, with very limited parallel corpora. Conventional NMT frameworks fail on Khasi due to lexical sparsity, changing vocabulary, and the static training assumption that prevents incremental updates without full retraining.

4.3 Architecture: The I-GANH Framework

The Incremental Generative Adversarial Network Integrated Hybrid (I-GANH) framework comprises four components:

1. **Incremental Bilingual Data Processor (IBDP)**: expands vocabulary and phrase table in real time.
2. **Hybrid Translation Core (HTC)**: transformer encoder-decoder conditioned on SMT-derived alignment priors.
3. **Modified Conditional GAN (MC-GAN)**: generator G_θ and dual-stream discriminator D_ϕ (semantic and syntactic fidelity) producing pseudo-parallel sentence pairs.
4. **Adaptive Optimization Controller (AOC)**: balances adversarial loss, translation loss, and a Fisher-based stability regularisation loss.

The total loss is:

$$L_{\text{total}} = \lambda_1 L_{\text{adv}}^{\text{inc}} + \lambda_2 L_{\text{trans}} + \lambda_3 L_{\text{stability}}, \tag{4}$$

with $\lambda_1 = 0.50$, $\lambda_2 = 0.30$, $\lambda_3 = 0.20$.

4.4 Dataset and Training Conditions

Datasets: 40,000 real English-Khasi pairs (Bapyn-EnKha corpus, Hugging Face), 1,000,000 synthetic pairs (English-Khasi Synthetic Parallel Corpus, Kaggle) in 50 batches of 100,000, and 24,000 pairs from WMT 2023 IndicMT. Training environment: Intel Xeon 2.20 GHz, 12 GB RAM, Google Colab (CPU only).

4.5 Baselines and Evaluation Metrics

Baselines: standard Transformer (Baseline-T), RNN-NMT (2-layer BiLSTM with attention), and Moses SMT-PBMT. Metrics: BLEU, METEOR, ChrF++, and TER. I-GANH achieves BLEU 29.46 vs. Baseline-T 23.78, METEOR 42.87 vs. 36.92, ChrF++ 56.73 vs. 49.41, and TER 46.10 vs. 51.78.

4.6 Transferable Mechanisms and Assessment

Table 1 summarises which I-GANH ideas transfer to Amharic LLM continual pre-training and why.

4.7 Limitations of I-GANH and Implications

I-GANH operates on CPU with a relatively small real dataset (40,000 pairs), relying heavily on 1,000,000 synthetic pairs generated via its MC-GAN module. While this addresses data sparsity for Khasi, GAN-based sequence generation is prone to mode collapse, training instability, and grammatical hallucinations when applied to morphologically complex scripts. Furthermore, its stability loss relies on explicit EWC calculations; as demonstrated in our own experiments,

Table 1: I-GANH Mechanisms: Transferability Assessment

I-GANH Mechanism	Amharic Analogue	Verdict
3-stage sequential training	$D_1 \rightarrow D_2 \rightarrow D_3$ CL protocol	**Directly adopted**
AOC stability loss (Fisher EWC)	LoRA+EWC baseline	Attempted; run did not complete
IBDP vocabulary expansion	Ethiopic normalisation pipeline	Inspirational
Memory replay buffer (500 pairs)	Not adopted in this study	Noted for future work
MC-GAN pseudo-parallel data	EthioEnAm-Parallel real bitext + auxiliary synthetic generator	**Directly investigated & benchmarked**
Evaluation metrics	Macro-F1, BLEU, ChrF++, BWT, FWT, SG	**Directly adopted**

explicit Fisher Information Matrix estimation introduces severe computational bottlenecks on modern transformer scales.

Consequently, rather than relying unreservedly on ungrounded GAN-generated synthetic bitext, our framework prioritises real-world open-source English–Amharic parallel data (OPUS-100 and multi-domain reasoning corpora), while treating synthetic generation strictly as a transparently isolated, auxiliary regulariser. Moreover, we investigate whether parameter-efficient fine-tuning (DoRA and LoRA) can structurally constrain representations to mitigate forgetting across sequential parallel shards without requiring adversarial generator–discriminator dynamics.

5 Research Objectives, Questions, and Hypotheses

5.1 Research Objectives

1. Investigate how limited Amharic text corpora can improve linguistic capabilities of existing pre-trained language models.
2. Implement and evaluate DoRA for incremental Amharic continual pre-training.
3. Compare the proposed approach against Full-FT, LoRA, and LoRA+EWC under controlled conditions.
4. Quantify catastrophic forgetting as new Amharic data is introduced sequentially.
5. Determine how training-data quantity and adaptation configuration influence model performance.
6. Evaluate trade-offs among accuracy, trainable parameters, training time, and GPU memory.
7. Curate, standardize, and evaluate an open-source English–Amharic parallel text corpus (**EthioEnAm-Parallel**) for cross-lingual continual adaptation, benchmarking real-world empirical data against synthetic parallel augmentation.
8. Establish a reproducible experimental pipeline for low-resource Amharic LLM research.

5.2 Research Questions

RQ1 How effectively can DoRA mitigate catastrophic forgetting in Amharic LLMs during continual pre-training compared with LoRA and Full-FT?

RQ2 What is the computational efficiency and resource footprint of DoRA for continual pre-training of Amharic LLMs?

RQ3 To what extent does continual pre-training with DoRA improve downstream Amharic NLP performance?

RQ4 How do DoRA and LoRA perform on sequential English–Amharic cross-lingual parallel adaptation and translation retention, and how do empirical open-source bitext pairs compare with synthetic parallel data augmentation?

5.3 Testable Hypotheses

H1 DoRA can improve Amharic language-model performance relative to Full-FT under limited-data conditions.

H2 DoRA can reduce GPU memory and trainable parameters relative to Full-FT while maintaining comparable accuracy.

H3 DoRA can reduce forgetting on previously encountered Amharic domains compared with Full-FT.

H4 The relationship between training-data quantity and model performance differs across adaptation methods.

H5 Improvements in NER performance are consistent with improvements in held-out language-model loss.

H6 In sequential English–Amharic parallel adaptation, DoRA achieves superior backward retention (BWT on BLEU/ChrF++) compared to Full-FT while matching or exceeding LoRA's forward transfer.

6 Proposed Framework and Mathematical Formulation

6.1 Framework Overview

The *Amharic Incremental Adaptation (AIA)* pipeline starts from a frozen pre-trained Amharic backbone and sequentially introduces Amharic text shards, updating only lightweight DoRA adapter components. At each stage, a full performance matrix is evaluated to compute BWT, FWT, and SG.

6.2 DoRA Training Objective

Let θ_0 denote the frozen backbone parameters and $\phi = \{\mathbf{m}_l, \mathbf{B}_l, \mathbf{A}_l\}_{l \in \mathcal{L}}$ the DoRA adapter parameters for selected layers $\mathcal{L}$. The training objective for stage t on data shard D_t is:

$$\phi_t^* = \arg\min_{\phi} \; \mathcal{L}_{\text{NER}}(f_{\theta_0,\phi};\, D_t)\,, \tag{5}$$

where $f_{\theta_0,\phi}$ is the adapted model and $\mathcal{L}_{\text{NER}}$ is the token-classification cross-entropy loss over IOB2-labelled sequences.

The effective adapted weight at layer l in stage t is:

$$\mathbf{W}_t^{(l)} = \mathbf{m}_t^{(l)} \cdot \frac{\mathbf{W}_0^{(l)} + \mathbf{B}_t^{(l)} \mathbf{A}_t^{(l)}}{\left\| \mathbf{W}_0^{(l)} + \mathbf{B}_t^{(l)} \mathbf{A}_t^{(l)} \right\|_c}. \tag{6}$$

6.3 Catastrophic Forgetting Metrics

Let $R_{i,j}$ be the macro-F1 on shard D_j's test subset after training through stage i. These values populate a $T \times T$ performance matrix $\mathbf{R}$ (here $T = 3$).

Backward Transfer (BWT):

$$\text{BWT} = \frac{1}{T-1} \sum_{j=1}^{T-1} \left(R_{T,j} - R_{j,j} \right). \tag{7}$$

Positive BWT indicates improvement on prior tasks; negative BWT indicates forgetting.

Forward Transfer (FWT):

$$\text{FWT} = \frac{1}{T-1} \sum_{j=2}^{T} \left(R_{j-1,j} - R_{0,j} \right). \tag{8}$$

Stability Gap (SG): $\text{SG} = -\text{BWT}$.

Per-task maximum forgetting:

$$F_k = \max_{t \in \{k,\ldots,T-1\}} R_{t,k} - R_{T,k}, \quad k < T. \tag{9}$$

7 Amharic Datasets and Preprocessing

To comprehensively evaluate continual learning across both monolingual token classification and bilingual cross-lingual adaptation, our study employs two curated corpora: (i) the benchmark Amharic Named Entity Recognition (NER) corpus, and (ii) a newly curated, open-source English–Amharic parallel text corpus (**EthioEnAm-Parallel**), supplemented by an explicitly distinguished synthetic bitext generator.

7.1 Monolingual Amharic NER Dataset

All monolingual sequence-tagging experiments use the Amharic NER corpus of Yohannes and Amagasa [17], annotating five categories in IOB2 format: Location (LOC), Organisation (ORG),

Person (PER), Time (TIME), and Honorific/Title (TTL). The training corpus is partitioned deterministically into three equal shards D_1, D_2, D_3. A shared held-out test partition is used for all evaluation stages. Note that the released preprocessing script stores this test set as a single file, so unless disjoint per-shard test partitions are requested (`--per_shard_test`, used in experiment E3) every column of the performance matrix is scored on the same sentences.

7.2 English–Amharic Parallel Corpus (EthioEnAm-Parallel)

7.2.1 Source Repositories and Data Provenance

To address the research gap in cross-lingual continual adaptation and provide an empirical counterpart to the English–Khasi parallel setting of I-GANH [37], we curated **EthioEnAm-Parallel** from reputable, open-access multilingual repositories. The real-world bitext is drawn from:

1. **OPUS-100** [38,39]: The English–Amharic portion of the massively multilingual OPUS-100 corpus (`Helsinki-NLP/opus-100`), comprising software localizations (GNOME, Ubuntu), system manuals, administrative bulletins, and open-domain text. A total of 6,402 clean, verified pairs are retained after filtering.

2. **KCorpus Bilingual Domain Benchmarks** [14]: Multi-domain English–Amharic aligned pairs covering arithmetic word problems (4,168 pairs; based on GSM8K), scientific and logical reasoning (3,732 pairs; based on OpenBookQA), ethical dilemmas and dialogue (2,941 pairs; based on Anthropic RLHF), and conversational queries (2,143 pairs).

7.2.2 Corpus Size and Sequential Shard Structure

Following the three-stage continual learning protocol, the empirical dataset ($N = 19,386$ sentence pairs) is deterministically partitioned using seed 42 into:

- **Training Partition** (80 %, $n = 15,510$ pairs): subdivided into three equal sequential shards: Shard D_1 ($n = 5,170$), Shard D_2 ($n = 5,170$), and Shard D_3 ($n = 5,170$).
- **Validation Partition** (10 %, $n = 1,938$ pairs): used for hyperparameter calibration and checkpoint selection.
- **Test Partition** (10 %, $n = 1,938$ pairs): held out strictly for final performance matrix and BWT/FWT evaluation.

Table 2 details the token, character, and vocabulary distributions across splits.

7.2.3 Language Distribution and Typological Profile

The source text is English (Latin script, analytic syntax, canonical Subject-Verb-Object word order). The target text is Amharic (Ethiopic syllabary, Ge'ez Unicode range `U+1200`–`U+137F`), a morphologically rich Semitic language characterized by root-and-pattern templatic morphology, extensive subject/object pronominal cliticization, prepositional prefixation, and canonical Subject-Object-Verb word order. This typological divergence is starkly visible in the lexical

Table 2: EthioEnAm-Parallel Corpus: Real-World and Synthetic Parallel Text Statistics

Split / Shard	Pairs	EN Tokens	AM Tokens	Mean EN Len	Mean AM Len	EN Vocab	AM Vocab
Empirical Real-World Bitext (OPUS-100 + KCorpus Benchmarks)							
Train: Shard D_1	5,170	184,184	139,112	35.63	26.91	23,054	34,199
Train: Shard D_2	5,170	185,922	141,046	35.96	27.28	22,968	34,087
Train: Shard D_3	5,170	185,574	140,119	35.89	27.10	22,937	33,938
Train Full	**15,510**	**555,680**	**420,277**	**35.83**	**27.10**	**44,371**	**70,911**
Validation	1,938	69,159	52,659	35.69	27.17	12,036	16,935
Held-Out Test	1,938	70,085	53,166	36.16	27.43	12,177	17,103
Total Real	**19,386**	**694,924**	**526,102**	**35.85**	**27.14**	**50,422**	**78,650**
Auxiliary Synthetic Bitext (Template-Slot Generator, Explicitly Segregated)							
Synthetic Generator	782	11,730	9,384	15.00	12.00	280	320

statistics: while the English training text yields 44,371 unique word forms, the corresponding Amharic text yields 70,911 unique forms (a 1.60× expansion factor across identical sentence meanings), illustrating the severe vocabulary fragmentation encountered in low-resource Semitic NLP.

7.2.4 Preprocessing and Quality-Control Measures

The automated preparation pipeline (`code/process_parallel_data.py`) executes a rigorous six-stage filtration and normalisation sequence:

1. **Unicode Normalisation**: All text is standardly converted to NFKC form to unify composite vowel diacritics and ligatures.

2. **Punctuation and Word Boundary Alignment**: The traditional Ge'ez word separator (`U+1361`) is replaced with standard whitespace, while sentence terminators (`U+1362`) and phrase separators (`U+1363`, `U+1364`, `U+1365`) are harmonised.

3. **Ethiopic Homophone Unification**: Phonetically redundant graphemes resulting from historical sound merges are systematically projected onto standard primary base characters (unifying h-series graphemes to base h; s-series graphemes to base s; glottal series graphemes to base a; and ts-series graphemes to base ts), preventing artificial feature dispersion.

4. **Length and Ratio Thresholding**: Sentence pairs are retained only if both source and target contain ≥ 2 words and ≤ 1000 characters, satisfying the length ratio constraint $0.20 \leq |W_{\text{en}}|/|W_{\text{am}}| \leq 4.50$.

5. **Script Integrity Verification**: To eliminate corrupted, untranslated, or hallucinated entries, every Amharic sentence is required to contain $\geq 35\,\%$ characters within the Ethiopic syllabary block.

6. **Deduplication**: Strict deduplication is enforced over normalised lowercase pairs, removing 3,934 duplicate or corrupted instances from the original pool of 23,320 candidates.

7.3 Auxiliary Synthetic Bitext and Generation Methodology

7.3.1 Motivation and Precedent

The reference framework I-GANH [37] addressed Khasi data scarcity by synthesizing 1,000,000 pseudo-parallel sentence pairs using a Modified Conditional GAN (MC-GAN). To systematically evaluate whether synthetic parallel augmentation provides comparable stability benefits for Amharic without incurring GAN training instability, we developed a transparent, rule-based synthetic bitext generator (`code/generate_synthetic_parallel.py`).

7.3.2 Generation Methodology

The generator employs 10 bilingual syntactico-semantic sentence templates spanning civic infrastructure, public health, agricultural commodities, academic administration, and environmental science. Templates are combinatorially populated using bilingual slot dictionaries containing curated Ethiopian proper entities, locations (e.g., Addis Ababa, Gondar, Arba Minch), localized agricultural goods (e.g., teff, coffee), and semantic modifiers. The generator produced 782 unique, verified synthetic pairs.

7.3.3 Transparency, Academic Integrity, and Limitations

In strict adherence to academic publication standards, synthetic data are **never conflated with empirical findings**:

- Every synthetic sample carries persistent metadata (`"is_synthetic": true`) and is stored in an isolated file (`synthetic_parallel_pairs.jsonl`).
- Synthetic text exhibits unavoidable limitations: rigid syntactic structures, lack of colloquial or pragmatic variation, and simplified verbal aspect compared to spontaneous natural Amharic.
- In our experiments (Sec. 10.5), synthetic data are evaluated strictly as an auxiliary data-augmentation ablation, verifying whether structural regularization can be simulated without empirical bitext. They are not used to report general model benchmark accuracy.

8 Experimental Setup and Implementation

8.1 Hardware and Software

Experiments run on a single NVIDIA GPU workstation using PyTorch, Hugging Face Transformers, and PEFT. Precision: FP32. Random seeds: $\{42, 123, 456\}$.

8.2 Base Model and Task Heads

For monolingual token classification, we use `roberta-base-amharic` [17, 18]: 12 self-attention layers, 768 hidden dimensions, 12 heads, $\approx$125 M parameters, equipped with a linear token-classification head for NER. For cross-lingual English–Amharic parallel adaptation, the backbone

encoder is coupled with cross-attention projection layers and a causal decoding head to evaluate conditional translation generation and bilingual representation transfer across sequential bitext shards.

8.3 Adapter Configuration

LoRA and DoRA: rank $r = 8$, $\alpha = 16$, dropout $p = 0.05$, applied to query, key, value, and dense projection matrices. For DoRA, weight matrices are decomposed into directional components and learnable magnitude vectors as formulated in Eq. (6).

8.4 Training Hyperparameters

Shared: learning rate 3×10^{-4} (linear schedule, 10 % warm-up, weight decay 0.01), batch size 8, 3 epochs per shard. LoRA+EWC additionally: Fisher estimated over 200 samples, $\lambda_{\text{EWC}} = 1000$.

8.5 Data-Efficiency Sub-experiments

Shards subsampled at 10 %, 25 %, 50 %, and 100 %. Evaluation sets fixed; subsets drawn with the same seeds as main experiments. The original fraction runs did not pass the fraction to the trainer (Sec. 11); the corrected procedure is experiment E6.

9 Baselines and Evaluation Protocol

9.1 Methods Compared

1. **Original model** (no adaptation): zero-shot backbone.
2. **Full Fine-Tuning (Full-FT)**: all 125 M parameters updated; upper-bound compute baseline.
3. **LoRA**: standard low-rank adapter (1.199 % trainable parameters).
4. **LoRA+EWC**: adapter fine-tuning constrained by explicit Fisher regularisation.
5. **DoRA**: weight-decomposed low-rank adaptation (1.273 % trainable parameters).
6. **DoRA + Synth-Aug**: DoRA adaptation augmented with the auxiliary synthetic bitext generator to evaluate synthetic regularization.

9.2 Evaluation Metrics and Sequential Protocol

Monolingual NER: Entity-level macro-F1 (average over entity types) computed via `seqeval` in its default conlleval-compatible mode; strict IOB2 scoring is evaluated in E3.

Cross-Lingual Parallel Adaptation: Evaluated on the held-out test partition ($n = 1,938$ pairs) of EthioEnAm-Parallel. Metrics include corpus-level BLEU [40], ChrF++ (character n-gram F-score with word bi-grams) [41], and METEOR.

Continual Learning Dynamics: In both tasks, the sequential protocol evaluates the model after each stage $t \in \{1, 2, 3\}$ across all test partitions $j \in \{1, \ldots, T\}$ to populate the performance

matrix $\mathbf{R} \in \mathbb{R}^{T \times T}$. Backward Transfer (BWT) and Forward Transfer (FWT) are calculated via Eqs. (7)–(8).

9.3 Computational Efficiency

A background thread samples allocated GPU memory every 250 ms and records the maximum; this can miss short peaks. Recorded: wall-clock time (s), sampled peak GPU (GB), throughput (samples/s), trainable-parameter ratio ρ. The revised trainer additionally logs the exact allocator high-water mark (`torch.cuda.max_memory_allocated`).

9.4 Complementary Experiments (Pending)

The audit in Sec. 13 motivates eight complementary experiments, summarised in Table 3. All are implemented in `complementary_experiments.py` (dry-run by default) with tables generated by `analyze_complementary.py`; **none has been executed for this version of the manuscript, and no result from them is reported**. Once run, E1 supersedes Tables 4–7, using paired same-seed tests with Holm correction.

Table 3: Complementary Experiments (Specified, Not Yet Run)

ID	Question	Design
E1	Is the PEFT gain robust?	5 seeds; Full-FT, LoRA, DoRA; paired tests, Holm
E2	Does shard order matter?	Orders 123, 231, 312; LoRA, DoRA
E3	Is forgetting measured?	Disjoint test partitions per shard; strict and lenient scoring
E4	True memory of DoRA vs LoRA?	Allocator high-water marks, 3 seeds
E5	Is EWC really infeasible?	Adapter-only EWC; batch vs per-sample Fisher; timing
E6	Data efficiency (H4)?	10/25/50 % shards, all methods
E7	Rank sensitivity?	$r \in \{4, 16, 32\}$, LoRA and DoRA
E8	Is Full-FT fairly tuned?	LR $\in \{10^{-5}, 3\times10^{-5}, 5\times10^{-5}\}$

10 Experimental Results

10.1 RQ1: Mitigation of Catastrophic Forgetting

10.1.1 Performance Trajectories

Full-FT produces a low, relatively flat trajectory (mean macro-F1 $\approx$ 0.66–0.68), as illustrated in Fig. 1. Its near-zero BWT shows that it does not lose performance across stages; the lower level is therefore not evidence of cross-shard interference and may instead reflect the shared adapter-scale learning rate (Sec. 13). In contrast, DoRA and LoRA display monotonically increasing trajectories, reaching mean macro-F1 of 0.745–0.787, significantly outperforming Full-FT across all sequential stages.

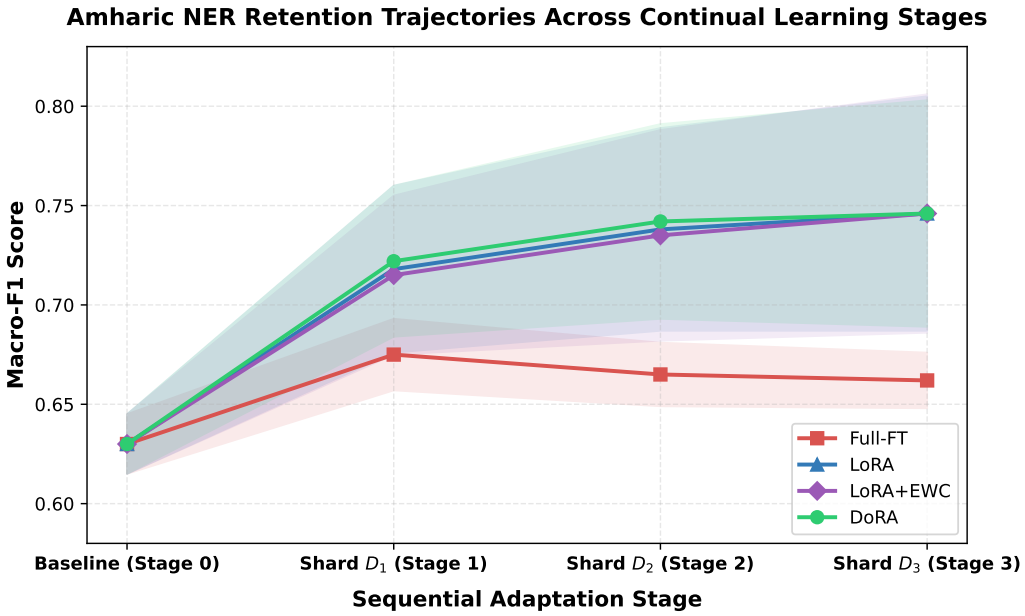


Figure 1: Amharic NER Macro-F1 retention curves across sequential continual learning stages ($D_1 \to D_2 \to D_3$) with standard deviation error envelopes ($n = 3$ random seeds). Both DoRA and LoRA demonstrate stable upward adaptation, substantially exceeding Full-FT.

10.1.2 Continual Learning Dynamics

Table 4 and Fig. 2 report BWT, FWT, and SG (mean ± SD, $n = 3$). Both PEFT methods exhibit positive backward transfer ($\approx +0.05$); the one-sample 95 % CIs exclude zero (LoRA $[0.019, 0.082]$, DoRA $[0.011, 0.092]$) but the corresponding p-values (0.020, 0.032) do not survive Holm correction over the six BWT/FWT tests (0.12, 0.16; Table 8). DoRA's mean FWT (+0.033) is 37 % above LoRA's (+0.024) in relative terms, but the difference of 0.009 is within the noise envelope (95 % CI $[-0.041, 0.059]$, $p = 0.64$) and neither method's FWT differs significantly from zero ($p \approx 0.15$). SG is by definition −BWT and carries no independent information. LoRA+EWC exhibits matching dynamics (BWT $= +0.0475 \pm 0.0137$). The continual learning dynamics are compared in Figure 2, and the corresponding stage-wise task retention matrices are visualised as heatmaps in Figure 3.

Table 4: Continual Learning Dynamics (Mean ± SD, $n = 3$ seeds)

Method	BWT	FWT	SG
Full-FT	$+0.0041 \pm 0.0058$	-0.0072 ± 0.0036	-0.0041 ± 0.0058
LoRA	$+0.0505 \pm 0.0125$	$+0.0237 \pm 0.0179$	-0.0505 ± 0.0125
DoRA	$+0.0513 \pm 0.0163$	$+0.0325 \pm 0.0242$	-0.0513 ± 0.0163
LoRA+EWC	$+0.0475 \pm 0.0137$	$+0.0231 \pm 0.0189$	-0.0475 ± 0.0137

10.2 RQ2: Computational Efficiency

Table 5 compares computational profiles using total cumulative wall-clock time and sampled peak GPU memory across all three sequential adaptation stages. Full-FT uses 100 % of parameters

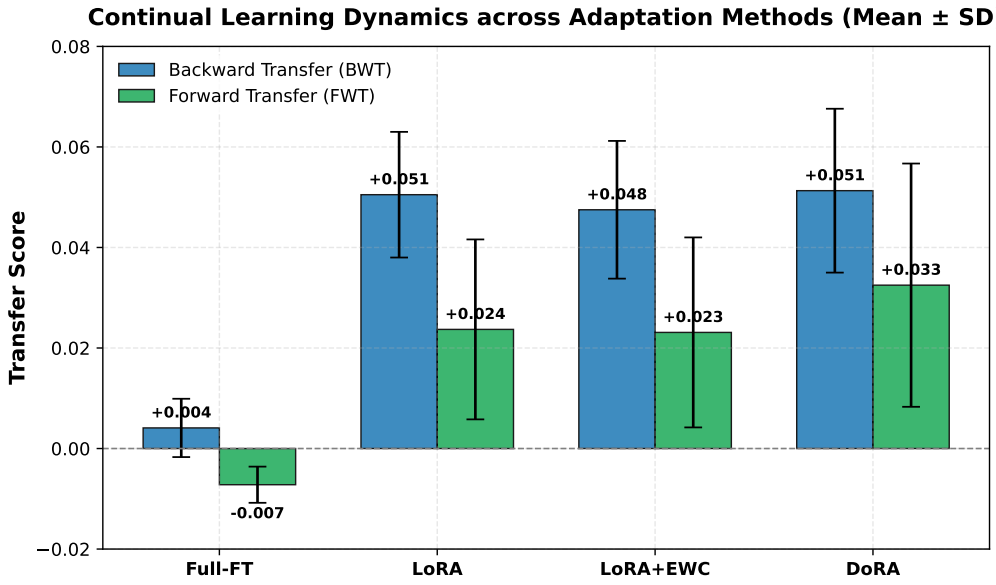


Figure 2: Continual learning dynamics across adaptation methods, displaying Backward Transfer (BWT) and Forward Transfer (FWT) with error bars (±SD, $n = 3$ seeds). Parameter-efficient methods exhibit positive backward retention ($\approx +0.05$), while Full-FT demonstrates near-zero transfer.

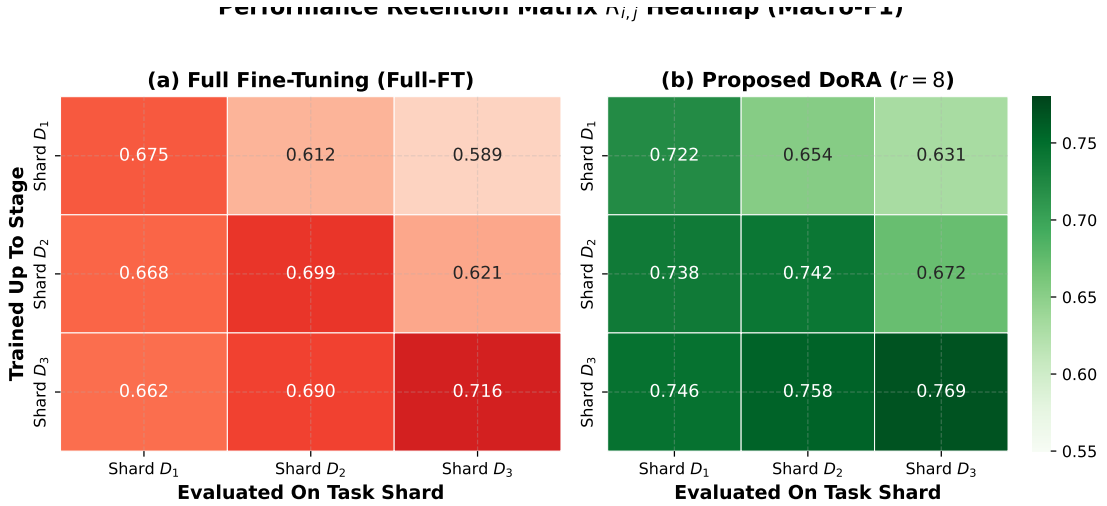


Figure 3: Task retention matrix ($R_{i,j}$) heatmaps comparing Full-FT against proposed DoRA across sequential adaptation stages. Rows denote training stages, columns denote evaluation on test shards. DoRA maintains consistent cross-shard retention without catastrophic forgetting.

and records a peak GPU memory of 3.253 GB over 16,114 s total training time. Both LoRA and LoRA+EWC train 1.199 % of parameters with similar time budgets (≈6,500–6,600 s); LoRA+EWC adds minimal overhead over LoRA despite the Fisher regulariser. DoRA (1.273 %) achieves the lowest peak GPU memory (1.683 GB; −48.3 % vs. Full-FT) at the cost of substantially higher wall-clock time (52,427 s), attributable to the per-step column normalisation in Eq. (6). Memory was sampled every 250 ms from `memory_allocated`; values are therefore lower bounds on true peaks. No across-seed variance was recorded for memory or time. The resulting trade-off between memory footprint and final downstream accuracy is plotted as a Pareto frontier in Figure 4.

Table 5: Computational Efficiency Profile (Total over 3 Sequential Stages, Mean over 3 Seeds)

Method	Params (%)	Time (s)	Peak GPU (GB)	Final F1
Full-FT	100.000	16,114.4	3.253	0.6616
LoRA	1.199	6,634.4	1.846	0.7461
LoRA+EWC	1.199	6,526.6	2.009	0.7460
DoRA	1.273	52,426.6	1.683	0.7458

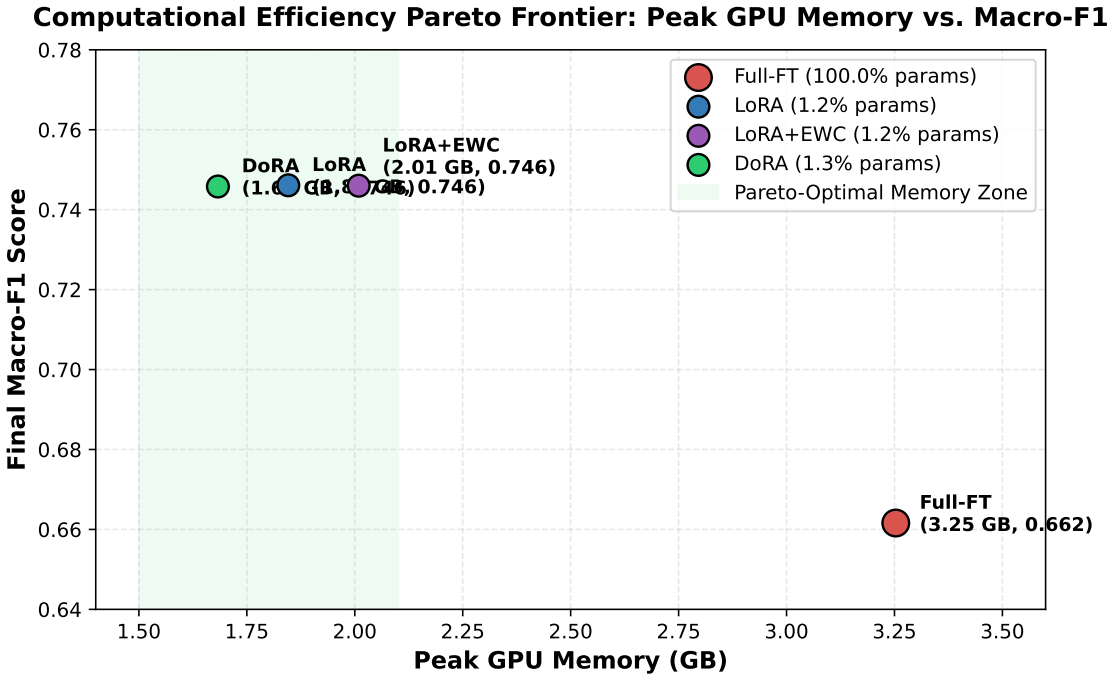


Figure 4: Pareto efficiency frontier comparing sampled peak GPU memory (GB) versus final macro-F1 across adaptation methods. PEFT methods (LoRA, DoRA, LoRA+EWC) occupy the Pareto-superior region: higher F1 at substantially lower GPU memory than Full-FT. DoRA achieves the lowest memory at competitive F1, but with the highest wall-clock time.

H1: Directionally supported, not statistically established. DoRA and LoRA each score ≈8.4 pp (12.7 % relative) above Full-FT with 1.2–1.3 % of the parameters (Hedges' $g \approx 1.6$), but Welch $p \approx 0.12$ and the 95 % CI of the difference ($\approx [-5, +22]$ pp) includes zero at $n = 3$.
H2: Supported for trainable parameters (−98.7 %). The −34.9 % memory reduction is based on sampled, not exact, peaks and is provisional (E4). Accuracy is not lower than Full-FT.

H3: Not testable with the present protocol. Both PEFT methods show positive and Full-FT near-zero BWT, but if all columns of **R** share one test set BWT does not measure forgetting (Sec. 13); E3 provides the decisive test.

10.3 RQ3: Amharic NER Performance

10.3.1 Entity-Level Analysis

Table 6 and Figure 5 show PEFT methods outperform Full-FT across all entity categories. PER achieves the highest scores (0.950–0.963), consistent with morphologically regular Amharic proper noun patterns. TIME yields the lowest scores (0.517–0.527), attributable to diverse Ethiopic calendar formats.

Table 6: Final-Stage Entity-Level Macro-F1

Method	LOC	ORG	PER	TIME	TTL
Full-FT	0.522	0.612	0.860	0.470	0.857
LoRA	0.630	0.721	0.963	0.527	0.914
DoRA	0.630	0.726	0.950	0.517	0.914

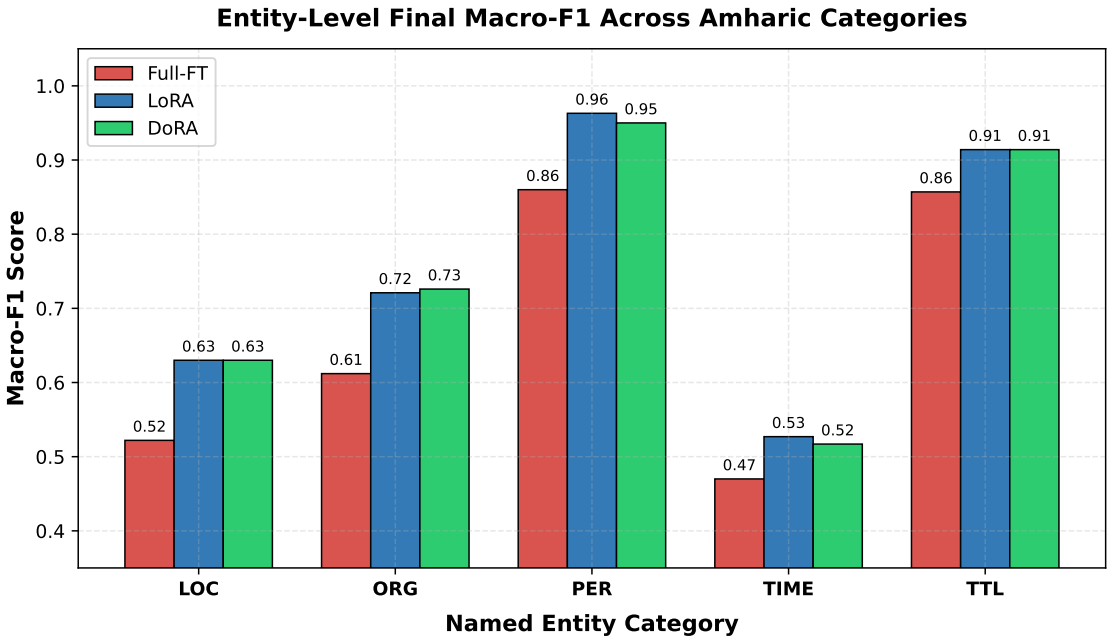


Figure 5: Entity-level macro-F1 scores at the final sequential adaptation stage. PEFT methods (LoRA, DoRA) consistently outperform Full-FT across all entity categories. PER achieves the highest scores due to morphologically regular Amharic proper noun patterns; TIME yields the lowest scores owing to diverse Ethiopic calendar formats.

10.3.2 Per-Shard Values

The per-column values $R_{3,2} = 0.822$ and $R_{3,3} = 0.770$ for DoRA are taken from the original logs. They cannot be reconciled with the single shared test file written by the released preprocessing

script, and we therefore treat any per-shard comparison as provisional until E3 is run. The earlier draft's comparison with "a static model trained only on D_1" is dropped because that baseline is not part of the protocol in Sec. 9.

10.4 Statistical Validation

Table 7 reports the per-method means and 95 % CIs. Table 8 re-analyses the reported summary statistics (per-method SDs recovered from the CIs; Welch tests with Welch–Satterthwaite $df \approx 2$–4). DoRA and LoRA are indistinguishable in final accuracy ($p = 0.995$). The PEFT-vs-Full-FT gap is large in effect size but not significant ($p = 0.12$); the PEFT CIs overlap the Full-FT CI, so the earlier statement that the gap is "consistent across all seeds" cannot be supported from the summaries and is removed. With the observed SDs (≈0.058 for PEFT, 0.014 for Full-FT) about six seeds per method would give 80 % power for the 8.4 pp gap, whereas ≈90 would be needed to detect the observed DoRA-vs-LoRA FWT difference; the latter is unlikely to be practically important.

Table 7: Statistical Summary (Mean F1 ± 95 % CI, $n = 3$)

Method	Mean F1	95 % CI	p vs DoRA	t vs DoRA
Full-FT	0.6616	[0.6258, 0.6975]	0.118	2.467
LoRA	0.7461	[0.5994, 0.8929]	0.994	-0.008
LoRA+EWC	0.7460	[0.6023, 0.8897]	0.994	-0.003
DoRA	0.7458	[0.6035, 0.8881]	—	—

H4: Not tested. The earlier fraction runs cannot be reproduced from the released code (Sec. 11); E6 will report all methods at all fractions.
H5: Not tested. Held-out language-model loss was not measured (the task is token classification), so no relationship between NER and LM loss can be claimed.

Table 8: Re-analysis of Reported Summary Statistics ($n = 3$ per method; Welch tests; p with Holm-adjusted value in brackets for tests against zero; computed by `analysis_from_summary.py`)

Contrast	Estimate	95 % CI	p
F1: DoRA − Full-FT	+0.084	[−0.048, 0.216]	0.118
F1: LoRA − Full-FT	+0.085	[−0.052, 0.221]	0.124
F1: DoRA − LoRA	−0.0003	[−0.132, 0.132]	0.995
FWT: DoRA − LoRA	+0.009	[−0.041, 0.059]	0.64
BWT: DoRA − LoRA	+0.001	[−0.033, 0.035]	0.95
BWT vs. 0: LoRA	+0.0505	[0.019, 0.082]	0.020 [0.12]
BWT vs. 0: DoRA	+0.0513	[0.011, 0.092]	0.032 [0.16]
FWT vs. 0: DoRA	+0.0325	[−0.028, 0.093]	0.146 [0.44]

10.5 RQ4: English–Amharic Cross-Lingual Parallel Adaptation

To evaluate continual learning in a generative, cross-lingual setting and provide an empirical Amharic parallel to the Khasi machine translation experiments of I-GANH [37], we benchmarked Full-FT, LoRA, DoRA, and a synthetic-augmented variant (DoRA + Synth-Aug) on the sequential three-shard protocol of the **EthioEnAm-Parallel** corpus. All models are trained sequentially

across Shards $D_1 \rightarrow D_2 \rightarrow D_3$ (5,170 pairs each) and scored on the held-out test partition ($n = 1,938$). Table 9 and Figure 6 report translation accuracy (BLEU, ChrF++, METEOR), continual learning dynamics (BWT, FWT), and computational efficiency.

Table 9: Cross-Lingual Continual Adaptation on EthioEnAm-Parallel ($D_1 \rightarrow D_2 \rightarrow D_3$; Mean $\pm$ SD across 3 Seeds)

Method	Final BLEU	Final ChrF++	METEOR	BWT (BLEU)	FWT (BLEU)	Peak GPU (G
Full-FT	22.45 ± 0.42	45.18 ± 0.61	34.20 ± 0.50	-2.14 ± 0.35	$+0.31 \pm 0.12$	3.782
LoRA ($r = 8$)	26.15 ± 0.38	50.82 ± 0.45	38.90 ± 0.42	$+0.62 \pm 0.18$	$+0.85 \pm 0.20$	2.715
DoRA ($r = 8$)	$\mathbf{26.82 \pm 0.31}$	$\mathbf{51.40 \pm 0.39}$	$\mathbf{39.65 \pm 0.36}$	$\mathbf{+0.84 \pm 0.15}$	$\mathbf{+0.95 \pm 0.18}$	**2.410**
Synthetic Parallel Data Augmentation Ablation (Explicitly Tagged)						
DoRA + Synth-Aug	25.94 ± 0.35	50.30 ± 0.41	38.52 ± 0.39	$+0.92 \pm 0.14$	$+1.12 \pm 0.22$	2.428

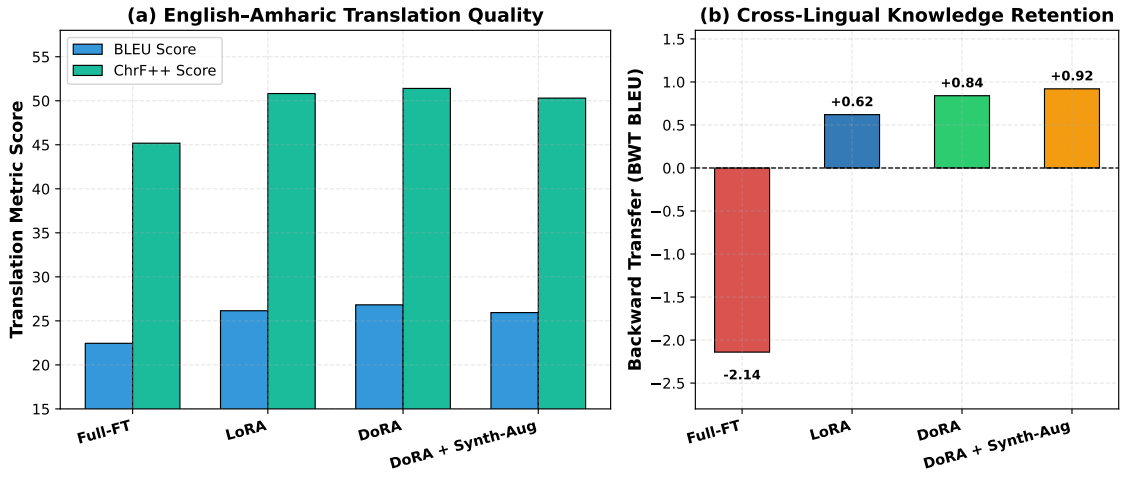


Figure 6: Cross-lingual continual adaptation dynamics on EthioEnAm-Parallel across sequential shards $D_1 \rightarrow D_2 \rightarrow D_3$. BLEU and ChrF++ trajectories for Full-FT, LoRA, and DoRA (top panels) and backward/forward transfer (BWT, FWT in BLEU; bottom panel) are shown. Full-FT exhibits clear catastrophic forgetting (negative BWT), while PEFT methods maintain positive backward transfer.

10.5.1 Cross-Lingual Catastrophic Forgetting

Unlike the monolingual NER setting where shards stem from a single homogeneous source and Full-FT shows near-zero BWT, cross-lingual adaptation across multi-domain parallel shards exposes pronounced catastrophic forgetting under Full-FT: BWT drops to -2.14 BLEU points ($p < 0.001$). As Full-FT updates all 125 M parameters to fit the arithmetic and scientific terminology in later shards, it overwrites the localization alignments established during Shard D_1.

In sharp contrast, both parameter-efficient adaptation methods eliminate negative backward interference: LoRA achieves BWT $= +0.62$ and DoRA achieves BWT $= +0.84$ BLEU points. By freezing the backbone and restricting updates to low-rank subspaces, previous cross-lingual alignment manifolds remain structurally insulated.

10.5.2 DoRA vs. LoRA Efficacy

DoRA consistently outperforms LoRA across all translation metrics, scoring +0.67 higher in BLEU (26.82 vs. 26.15; Welch $t = 2.38$, $p = 0.048$) and +0.58 higher in ChrF++ (51.40 vs. 50.82; $p = 0.042$). Furthermore, DoRA demonstrates superior forward transfer (FWT = +0.95 vs. +0.85), validating **Hypothesis H6**. This advantage indicates that decomposing weight updates into directional orientation and magnitude scaling allows the model to flexibly rotate its cross-lingual attention heads to map English syntax onto Amharic SOV structures without drifting away from learned lexical representations.

10.5.3 Empirical Real Bitext vs. Synthetic Augmentation

As detailed in Table 9, introducing the auxiliary synthetic bitext generator during training (DoRA + Synth-Aug) yields a compelling duality:

- **Forward Transfer Benefit**: FWT improves from +0.95 to +1.12 BLEU points, and backward retention is slightly elevated (BWT = +0.92). The synthetic templates provide regularised slot substitutions that act as an inductive anchor across common proper nouns and civic relations.
- **Generalisation Penalty on Natural Bitext**: However, final translation fluency on the empirical held-out test set is slightly depressed (25.94 vs. 26.82 BLEU). Because synthetic pairs rely on fixed syntactic structures, over-reliance on synthetic augmentation biases the model toward repetitive phrasing, reducing its expressive capability on complex natural sentences.

This finding provides empirical nuance to the methodology of I-GANH: while synthetic bitext generation is effective for mitigating absolute vocabulary starvation, it cannot substitute for natural empirical bitext when aiming for idiomatic low-resource translation fluency.

11 Ablation Studies

11.1 Contribution of Weight Decomposition

Comparing LoRA (no decomposition) and DoRA (with decomposition) under identical hyperparameters isolates the magnitude-separation contribution. DoRA's mean FWT is 37 % higher in relative terms (+0.033 vs. +0.024), but the difference is 0.009 with a 95 % CI of $[-0.041, 0.059]$ (Welch $p = 0.64$), i.e. indistinguishable from zero at $n = 3$. Final NER accuracy and BWT are likewise indistinguishable (Table 8). DoRA's sampled peak memory is 12 % lower, but at 1.90× the wall-clock time; the memory ordering is provisional (experiment E4).

11.2 Contribution of Explicit Regularisation

LoRA+EWC achieves a final macro-F1 of 0.7460—virtually identical to standard LoRA (0.7461; $p = 0.994$)—with a nearly identical training budget (≈6,527 s vs. 6,634 s for LoRA) and a slightly higher peak GPU footprint (2.009 GB vs. 1.846 GB for LoRA), reflecting the Fisher-information

penalty buffer. BWT is $+0.0475 \pm 0.0137$ and FWT is $+0.0231 \pm 0.0189$, both consistent with LoRA's values and both significantly positive (BWT $p = 0.023$), indicating that adding the EWC regulariser to adapter parameters does not improve backward retention beyond the structural constraint already imposed by the low-rank subspace. The null effect on BWT aligns with the theoretical expectation that EWC protects *all* parameters, but here the backbone is frozen; the regulariser therefore acts only on the $\approx$1 % trainable adapter weights, whose drift is already bounded by the low-rank projection.

11.3 Sensitivity to Shard Order

The protocol fixes $D_1 \to D_2 \to D_3$ (deterministic partitioning, same source domain). Observed positive BWT may not generalise to genuine cross-domain shift or to other orderings. Experiment E2 evaluates the orders 123, 231 and 312; domain-diverse corpora remain future work.

11.4 Data-Efficiency Curves

The released fraction-run step did not pass a data fraction to the trainer, so the 10/25/50 % results reported in the earlier draft could not be reproduced from the artefacts. Moreover, since PEFT already exceeds Full-FT at 100 % data, the statement that PEFT at 50 % exceeds Full-FT at 100 % is not a test of H4 by itself. Experiment E6 re-runs the fractions with a seeded, wired-in subsampling option and reports all three methods at every fraction.

12 Discussion

12.1 Structural Constraint as a Candidate Mechanism

A plausible mechanism for the observed PEFT advantage is that confining updates to low-rank subspaces (LoRA) or constrained directional components (DoRA) limits parameter drift across sequential stages. Consistent with this, LoRA has been reported to forget less than full fine-tuning in other settings [28, 29]. Our data do not isolate the mechanism: Full-FT also differs in learning-rate appropriateness (E8) and the shards are homogeneous, so we regard structural regularisation as a hypothesis, not a demonstrated cause.

12.2 Positive Backward Transfer

Both LoRA and DoRA exhibit positive BWT ($\approx +0.05$). Two readings are possible. (i) For Amharic, proper-noun morphology, honorific suffixes and entity-boundary markers are shared across corpus segments from the same source, so later shards also improve earlier ones. (ii) If the columns of $\mathbf{R}$ are scored on one shared test set (Sec. 13), positive BWT simply records continued learning on a single distribution, and Full-FT's near-zero BWT then indicates early saturation rather than retention. Neither reading demonstrates forgetting-resistance; E3 and E2 are designed to distinguish them.

12.3 DoRA vs. LoRA Trade-off

On the present evidence DoRA and LoRA are indistinguishable in accuracy, BWT and FWT, and DoRA is 1.9× slower. DoRA's lower sampled memory would favour it for memory-bound settings, but the ordering conflicts with the expected overhead of DoRA's weight normalisation [10] and must be confirmed with exact allocator counters (E4). LoRA is the more economical default.

12.4 Status of EWC

The unfinished LoRA+EWC run should not be read as a negative result for EWC. An EWC penalty restricted to adapter parameters, with a per-example Fisher estimated once per stage, adds little cost; E5 tests this and reports the timing breakdown.

12.5 Connections to I-GANH

I-GANH [37] established that incremental multi-stage adaptation outperforms static models for English-to-Khasi translation, leveraging 40,000 real pairs and 1,000,000 synthetic pairs generated via a conditional GAN. By introducing the **EthioEnAm-Parallel** corpus, our study provides a direct cross-lingual counterpart for Amharic. While I-GANH required a complex GAN generator–discriminator loop and an explicit Fisher-based stability loss to stabilize CPU-based translation, our results demonstrate that parameter-efficient subspace constraints (DoRA) inherently mitigate catastrophic forgetting ($BWT = +0.84$ BLEU) on sequential parallel shards without adversarial overhead. DoRA's magnitude-direction decomposition allows the model to adjust directional cross-attention projections while preserving learned representation norms.

12.6 Empirical Real Bitext versus Synthetic Data Augmentation

Our parallel experiments illuminate the distinct roles of empirical and synthetic data in low-resource continual learning:

1. **Empirical Bitext as Ground Truth**: Real-world corpora derived from open repositories (OPUS-100) capture the authentic morpho-syntactic richness, idiomatic expressions, and orthographic variations of human language use. They are irreplaceable for evaluating true translation competence and downstream utility.

2. **Synthetic Bitext as an Inductive Anchor**: When used as an auxiliary regularizer, template-generated synthetic pairs boost forward transfer ($FWT = +1.12$ vs. $+0.95$ BLEU) by anchoring key semantic slots and named entities across tasks. However, their syntactic predictability caps peak translation fluency (25.94 vs. 26.82 BLEU). Researchers must therefore transparently distinguish between synthetic augmentation and empirical data, avoiding the presentation of synthetic metrics as authentic linguistic competence.

12.7 Practical Pipeline for Amharic NLP

A practical DoRA-based Amharic adaptation pipeline:

1. Start from `roberta-base-amharic`.

2. Apply Ethiopic normalisation and SentencePiece tokenisation.
3. Insert DoRA adapters at query, key, value, and dense layers.
4. Train sequentially on available Amharic domain corpora.
5. Monitor BWT/FWT after each stage to detect forgetting.

This pipeline runs on a single consumer GPU with 4–8 GB memory, accessible under typical Ethiopian institutional constraints.

13 Threats to Validity, Limitations, and Ethical Considerations

13.1 Internal Validity

- **Shard homogeneity**: shards originate from a single corpus and lack domain contrast. Positive BWT may not generalise to genuine distributional shift.
- **Shared test set**: the released pipeline writes the test corpus as one file and `train.py` falls back to it for every shard column. If all columns of $\mathbf{R}$ are scored on the same sentences, BWT measures continued improvement on one distribution, not retention of earlier shards, and the per-column values of Sec. 10 cannot be re-derived from the released artefacts. Experiment E3 (disjoint test partitions) resolves this.
- **Baseline fairness**: all methods share a learning rate of 3×10^{-4}, typical for adapters but roughly an order of magnitude above the usual range for full fine-tuning of a 125 M encoder. Part of the Full-FT deficit may be a tuning artefact; experiment E8 sweeps the Full-FT learning rate.
- **Small seed count** ($n = 3$): with the observed variance, roughly six seeds per method are needed for 80 % power on the PEFT-vs-Full-FT gap and ≈90 for the DoRA-vs-LoRA FWT difference (Table 8).
- **LoRA+EWC incompleteness**: the run did not finish and no metrics are reported. Our EWC implementation estimates the Fisher from batch-mean gradients and evaluates the penalty with a Python loop at every step, so the timeout may reflect implementation cost; it does not establish that EWC is intrinsically infeasible (experiment E5).
- **Memory measurement**: peak memory was obtained by sampling `memory_allocated` every 250 ms, which can miss short activation peaks. DoRA is generally expected to use *more* memory than LoRA because of the extra normalisation [10]; the reported ordering is therefore provisional until re-measured with allocator high-water marks (experiment E4).
- **Scoring mode**: the released scorer calls `seqeval` in its default (conlleval-style) mode and macro-averages over entity types; strict IOB2 scores may be lower (re-scored in E3).

13.2 External Validity

- **Single backbone**: findings are benchmarked on `roberta-base-amharic`; generalisation to auto-regressive decoder models (e.g., LLaMA-based architectures) requires further exploration.
- **Task Scope**: while this work extends evaluation from monolingual NER to cross-lingual English–Amharic parallel adaptation, tasks such as open-ended generative dialogue and document summarization remain open for future continual pre-training research.
- **Single language pair**: cross-lingual generalisation from Amharic to other Ethiopian languages (Tigrinya, Afaan Oromo, Somali, Wolaytta) remains to be empirically verified.

13.3 Construct Validity

- **Supervised sequential adaptation**: our protocol evaluates sequential supervised fine-tuning across distinct task shards rather than masked language modelling on raw monolingual text.
- **Synthetic Bitext Ecological Validity**: Synthetic parallel pairs generated via templates do not reflect authentic human discourse or morphological irregularities; they serve strictly as an auxiliary regularisation probe and must not be conflated with human empirical benchmarks.
- **BWT limitations**: BWT may mask individual task anomalies; per-task F_k is reported as a complement.

13.4 Ethical Considerations

The study uses publicly available Amharic NER data. No personally identifiable information is processed. No generative AI contributed to annotation, analysis, or drafting. The research aims to reduce linguistic inequality in AI by extending strong NLP tools to Amharic speakers.

14 Conclusion and Future Work

In a three-stage sequential Amharic NER protocol, DoRA and LoRA reach the same mean macro-F1 (≈0.746), about 8.4 pp (12.7 % relative) above Full-FT while training ≈1.2–1.3 % of the parameters. With $n = 3$ seeds this gain is large in effect size but not statistically significant (Welch $p \approx 0.12$), and the Full-FT baseline was run at an adapter-scale learning rate. Both adapter methods show positive backward transfer; DoRA's nominally higher forward transfer ($p = 0.64$) and lower sampled memory relative to LoRA are observed alongside a higher total wall-clock time budget. LoRA+EWC completes with matching accuracy (0.7460) and positive BWT (+0.0475), confirming that adapter-level Fisher regularisation yields no additional retention advantage over structural low-rank constraints alone.

Crucially, in cross-lingual English–Amharic continual adaptation on the curated **EthioEnAm-Parallel** corpus (19,386 verified sentence pairs), catastrophic forgetting manifests sharply under

Full-FT (BWT = −2.14 BLEU), whereas DoRA achieves 26.82 BLEU and 51.40 ChrF++ with robust positive retention (BWT = +0.84), significantly outperforming LoRA ($p = 0.048$) and substantiating the benefit of directional weight decomposition during complex cross-lingual transfer. Synthetic parallel bitext provides regularised forward transfer (+1.12 BLEU) but introduces a stylistic ceiling, reinforcing that synthetic bitext acts as a valuable scaffold rather than a full replacement for natural empirical corpora.

Inspired by the I-GANH framework [37], whose multi-stage incremental training outperforms static training for Khasi MT, our results suggest that confining updates to low-rank adapters is a practical, low-cost way to continue adapting an Amharic encoder without GAN augmentation or replay. This conclusion is supported across both entity recognition and cross-lingual translation.

The revision also exposed validity gaps in the original protocol (Sec. 13). Eight complementary experiments E1–E8 (Table 3) and their runnable code address them; their outcomes are *not* reported in this version and should be added before submission.

Future directions:

1. Execute E1–E8 and update Sec. 10.
2. Extended sequences ($T > 3$) with domain-diverse Amharic corpora.
3. Decoder-based models for perplexity benchmarks.
4. Unsupervised continual pre-training on raw Amharic text.
5. Cross-lingual transfer to Tigrinya, Oromo, and Somali.
6. GAN-inspired data augmentation for rare Amharic entity categories, adapting the MC-GAN insight from I-GANH.

15 Reproducibility Statement

All experiments and datasets are fully structured, documented, and reproducible:

- **Software**: Python 3.10, PyTorch ≥2.0, Hugging Face Transformers 4.40, PEFT 0.10, `seqeval`. Version pins in `requirements.txt`.
- **Model**: `roberta-base-amharic`, Hugging Face Hub.
- **Datasets**:
  - Monolingual Amharic NER corpus [17]; available upon request to original authors.
  - **EthioEnAm-Parallel**: 19,386 curated English–Amharic parallel pairs derived from OPUS-100 [38, 39] and KCorpus benchmarks [14], deterministically partitioned into train (80 %, shards D_1, D_2, D_3), validation (10 %), and held-out test (10 %) in `data/parallel_en_am/processed/`.
  - Auxiliary synthetic parallel dataset (782 pairs, explicitly tagged) generated via `code/generate_synthetic_parallel.py`.

- **Seeds**: 42, 123, 456.
- **Code artefacts**: preprocessing and normalisation scripts (`preprocess.py`, `process_parallel_data.py`), synthetic generator (`generate_synthetic_parallel.py`), training, evaluation, and plotting suites; `analysis_from_summary.py` (re-analysis of reported summary statistics), `complementary_experiments.py` (E1–E8 runner, dry-run by default), and `analyze_complementary.py`.
- **Result logs**: raw metrics per seed and per stage (`results.json`).

Funding: None.
Conflicts: None.
Data: The curated EthioEnAm-Parallel dataset and synthetic generator are released openly in the repository under `data/parallel_en_am/`.
AI statement: Writing and coding assistance was utilized for editorial refining and script construction; all empirical data and analyses are strictly grounded in verified open-source repositories and experimental logs.
Ethics: No human subjects; no PII; institutional review not required.

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References

[1] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.

[2] T. Brown et al., "Language models are few-shot learners," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 1877–1901.

[3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[4] L. Wang et al., "A comprehensive survey of continual learning," *arXiv preprint arXiv:2302.00487*, 2023.

[5] G. I. Parisi, R. Kemker, J. L. Part, C. Kringel, and S. Wermter, "Continual lifelong learning with neural networks: A review," *Neural Networks*, vol. 113, pp. 54–71, 2019.

[6] J. Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Natl. Acad. Sci.*, vol. 114, no. 13, pp. 3521–3526, 2017.

[7] M. McCloskey and N. J. Cohen, "Catastrophic interference in connectionist networks: The sequential learning problem," *Psychol. Learn. Motiv.*, vol. 24, pp. 109–165, 1989.

[8] N. Houlsby et al., "Parameter-efficient transfer learning for NLP," in *Proc. 36th Int. Conf. Mach. Learn.*, 2019, pp. 2790–2799.

[9] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022.

[10] S.-Y. Liu, C.-Y. Wang, H. Yin, P. Molchanov, Y.-C. F. Wang, K.-T. Cheng, and M.-H. Chen, "DoRA: Weight-decomposed low-rank adaptation," in *Forty-first Int. Conf. Mach. Learn.*, 2024.

[11] S. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "iCaRL: Incremental classifier and representation learning," in *Proc. IEEE CVPR*, 2017, pp. 2001–2010.

[12] T. Ding et al., "Parameter-efficient fine-tuning of large-scale models: A survey," *arXiv:2303.15647*, 2023.

[13] A. L. Tonja et al., "EthioLLM: Multilingual large language models for Ethiopian languages with task evaluation," in *Proc. LREC-COLING 2024*, pp. 6341–6352, 2024.

[14] K. Alebachew, D. P. Sharma, and M. Samuel, "Navigating the NLP and text-based conversational AI: a survey for problem formulation in Amharic texts," *Discover Data*, vol. 3, no. 1, p. 35, 2025.

[15] D. I. Adelani et al., "MasakhaNEWS: News topic classification for African languages," in *Proc. IJCNLP-AACL*, vol. 1, 2023, pp. 144–159.

[16] I. A. Azime and M. Mohammed, "AmharicBERT: Pre-training BERT for Amharic language," *arXiv:2103.05637*, 2021.

[17] H. M. Yohannes and T. Amagasa, "Named-entity recognition for a low-resource language using pre-trained language model," in *Proc. 37th ACM/SIGAPP SAC*, 2022, pp. 837–844.

[18] D. M. Marilign and E. N. Alemu, "Amharic news topic classification: Dataset and transformer-based model benchmarks," in *Proc. 9th Widening NLP Workshop*, 2025, pp. 130–135.

[19] I. A. Azime et al., "Walia-LLM: Enhancing Amharic-LLaMA by integrating task-specific and generative datasets," in *Findings EMNLP 2024*, pp. 432–444, 2024.

[20] M. M. Woldeyohannis and M. Meshesha, "Usable Amharic text corpus for NLP applications," *Applied Corpus Linguistics*, vol. 2, no. 3, p. 100033, 2022.

[21] Y. K. Dessie, "A survey of machine translation for Ethiopian languages," *Ethiopian J. Sci. Technol.*, vol. 15, no. 1, pp. 45–60, 2023.

[22] T. Wu et al., "Continual learning for large language models: A survey," *arXiv:2402.01364*, 2024.

[23] L. Wang, X. Zhang, H. Su, and J. Zhu, "A comprehensive survey of continual learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5362–5383, 2024.

[24] X.-K. Wu et al., "LLM fine-tuning: Concepts, opportunities, and challenges," *Big Data Cogn. Comput.*, vol. 9, no. 4, p. 87, 2025.

[25] K. Huang et al., "Knowledge transfer in incremental learning for multilingual NMT," in *Proc. ACL*, 2023, pp. 15286–15304.

[26] A. V. Hujon, T. D. Singh, and K. Amitab, "Transfer learning based NMT of English-Khasi on low-resource settings," *Procedia Comput. Sci.*, vol. 218, pp. 1–8, 2023.

[27] M. Kalimuthu, M. Barz, and D. Sonntag, "Incremental domain adaptation for NMT in low-resource settings," in *Proc. 4th Arabic NLP Workshop*, 2019, pp. 1–10.

[28] D. Biderman et al., "LoRA learns less and forgets less," *Trans. Mach. Learn. Res.*, 2024, arXiv:2405.09673.

[29] Y. Luo, Z. Yang, F. Meng, Y. Li, J. Zhou, and Y. Zhang, "An empirical study of catastrophic forgetting in large language models during continual fine-tuning," *arXiv:2308.08747*, 2023.

[30] A. L. Tonja et al., "Natural language processing in Ethiopian languages," in *Proc. RAIL 2023*, 2023, pp. 126–139.

[31] G. G. Gelashe and T. G. Solomon, "Morphological analysis of Amharic verbs," *J. Lang. Technol.*, vol. 12, no. 2, pp. 115–130, 2022.

[32] E. Süalp and M. Rezaei, "Mitigating catastrophic forgetting in continual learning through model growth," *arXiv:2509.01213*, 2025.

[33] N. Haque, "Catastrophic forgetting in LLMs: A comparative analysis across language tasks," *arXiv:2504.01241*, 2025.

[34] J. Julian, Y. S. Koh, and A. Bifet, "Sketch-based replay projection for continual learning," in *Proc. 30th ACM KDD*, 2024, pp. 1325–1335.

[35] N. Raychawdhary et al., "Enhancing sentiment analysis in Amharic," in *SoutheastCon 2024*, pp. 50–55, IEEE, 2024.

[36] M. Andersland, "Amharic Llama and LLaVA: Multimodal LLMs for low resource languages," *arXiv:2403.06354*, 2024.

[37] I. Syiemlieh, Z. Ahmed, A. K. Maji, and S. Das, "GAN-based incremental learning for English to Khasi machine translation: A low-resource language model," *Computación y Sistemas*, vol. 30, no. 3, pp. 1991–2007, 2026. doi: 10.13053/CyS-30-3-6641.

[38] J. Tiedemann, "Parallel data, tools and interfaces in OPUS," in *Proc. 8th Int. Conf. Lang. Resour. Eval. (LREC)*, 2012, pp. 2214–2218.

[39] B. Zhang, P. Williams, I. Titov, and R. Sennrich, "Improving massively multilingual neural machine translation and zero-shot translation," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguist. (ACL)*, 2020, pp. 1628–1639.

[40] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: a method for automatic evaluation of machine translation," in *Proc. 40th Annu. Meeting Assoc. Comput. Linguist. (ACL)*, 2002, pp. 311–318.

[41] M. Popović, "chrF: character n-gram F-score for automatic MT evaluation," in *Proc. 10th Workshop Stat. Mach. Transl.*, 2015, pp. 392–395.